

C7.2 Energy changes in chemistry

Why are there energy changes during chemical reactions?

1. understand the terms exothermic and endothermic;
2. use simple energy level diagrams to represent exothermic and endothermic changes;
3. understand that energy is needed to break chemical bonds and that energy is given out when chemical bonds form;
4. **use data on the energy needed to break given covalent bonds to estimate the overall energy change in simple examples (for example the formation of steam or hydrogen halides from their elements) laS 3.1, 3.3, 3.4;**
5. understand the term activation energy in terms of the energy needed to break bonds to start a reaction.